

Smartphone Sales Analysis

Sr no.	Title	Subsections	Page No.
1	Introduction	-Purpose -Scope	2
2	System Overview	-System Architecture -Flow Chart -System Components	2
3	Data Management	-Data Sourcing -ETL Process	3
4	Data Modelling	-Entity-Relationship Diagram (ERD) -Schema Design	4
5	Visualization & User Interface (Implement Strategy)	-Layout -Features & Functionalities	4
6	Technical Requirements	-Software & Tools -Hardware Requirements	5
7	Milestones & Timeline	-	5
8	Analysis Plan	-	6
9	Conclusion & Future recommendations		6

Introduction:

The **Smartphone Sales Analysis** Dashboard in Power BI helps businesses track sales trends, understand customer preferences, and refine pricing strategies. It provides insights into top-selling models, sales performance, and profitability, allowing businesses to make data-driven decisions to boost revenue.

❖ Purpose:

The main goal is to help businesses make smart decisions by analyzing sales trends, discounts, profits, and competitor prices. This dashboard improves market positioning and pricing strategies.

❖ Scope:

- **Sales & Discounts** – See how discounts affect sales.
- **Market & Competitors** – Compare prices and market share.
- **Financial Overview** – Monitor revenue and profit trends.
- **Power BI & Data Analysis** – Use DAX, Power Query, and visuals for insights.

System Overview:

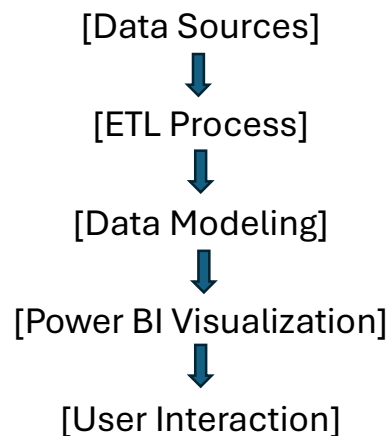
The system collects, analyzes, and displays smartphone sales data, helping businesses make quick, informed decisions. It combines different data sources, processes the data, and presents insights using interactive visuals.

❖ System Architecture:

The system is built with:

- **Data Collection** – Gathering sales data from different sources.
- **ETL Process** – Extracting, transforming, and loading data into Power BI.
- **Data Modeling** – Organizing data for easy analysis.
- **Visualization** – Displaying insights through Power BI dashboards.

❖ **Flow Chart:**



❖ **System components :**

- **Data Sources** – Sales records, product info, and price history.
- **Processing** – Power Query for data transformation.
- **Storage** – Power BI Data Model.
- **Visualization** – Power BI reports and dashboards.

Data Management :

The system uses organized methods to manage data, ensuring it remains accurate, reliable, and consistent. Data is first processed through ETL pipelines before being saved in an organized format.

❖ Data Sourcing:

- **Sales Transaction Data** – Includes details on the number of sales, revenue, and discounts.
- **Product Details** – Information on the model name, battery, camera, and chipset.
- **Store Performance Data** – Shows sales and revenue contribution for each store.

❖ ETL Process:

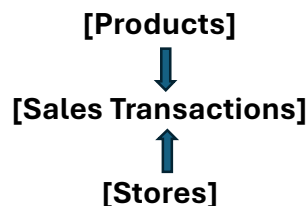
The ETL (Extract, Transform, Load) process includes:

- **Extract:** Data collected from multiple sources (CSV, databases).
- **Transform:** Cleaning, aggregating, and structuring data.
- **Load:** Storing processed data in Power BI.

Data Modelling

- **Fact Table:** Stores sales transactions.
- **Dimension Tables:** Stores product details, store locations, pricing information.

❖ ERD Diagram:



❖ Schema Design:

- **Fact_Sales** (SalesID, ModelID, StoreID, SalesAmount, Discount)
- **Dim_Products** (ModelID, Brand, Battery, Camera, Price)
- **Dim_Stores** (StoreID, Location, Region)

Visualization & User Interface(Implement Strategy)

- Interactive dashboards with slicers for filtering.
- Tooltips for additional insights.

❖ Layout:

1. Sales Overview

- 2. Discount Analysis
- 3. Financial Performance
- 4. Market Share Insights

❖ **Features & Functionalities:**

- Drill-through reports
- Customizable date filters
- KPIs for total revenue, profit, and sales growth

Technical Requirements

❖ **Software & Tools:**

- **Power BI** – Data visualization and analysis
- **SQL Server** – Data storage
- **Azure Synapse** – Data processing

❖ **Hardware Requirements:**

- **Processor:** Intel i5/i7 or higher
- **RAM:** 16GB+ for large datasets
- **Storage:** 500GB

Milestones and Timeline

Milestone	Task	Timeline
Data Collection	Gather sales data	Week 1
ETL Process	Clean and transform data	Week 2
Data Modeling	Define relationships	Week 3
Dashboard Development	Create visuals	Week 4
Testing & Deployment	Final review	Week 5

Analysis Plan

- Identify best-selling models.
- Compare sales trends across brands.
- Evaluate the impact of discount strategies.

Conclusion and future analysis:

This Smartphone Sales Analysis Dashboard enables businesses to make data-driven decisions. Future enhancements include:

- **AI-powered sales forecasting.**
- **Integration with external market trend analysis tools.**
- **Automated alerts for pricing optimization.**